

Pontifical Catholic University of São Paulo

Graduate Studies Program in

Intelligence Technologies and Digital Design (TIDD)

Fabiana Rotella Campaner

Evaluating visual detectors without ground truth

Data lineage and human review in helipad detection in satellite imagery of São Paulo

Research pre-proposal

Selection Process 1st/2027, Master's

São Paulo

2026

Contents

ABSTRACT **3**

1 RESEARCH QUESTION **3**

1.1 Context and problem 3

1.2 Central question and derived questions 5

2 STATE OF THE ART **6**

3 JUSTIFICATIONS **7**

4 OBJECTIVES **7**

5 HYPOTHESES **8**

6 THEORETICAL FRAMEWORK **8**

6.1 Evidence when there is no ground truth 8

7 METHODOLOGY **9**

8 TIMELINE **11**

PRELIMINARY BIBLIOGRAPHY **12**

Evaluating visual detectors without ground truth

Data lineage and human review in helipad detection in satellite imagery of São Paulo

Document: research pre-proposal, Selection Process 1st/2027, Master's.

Program: Intelligence Technologies and Digital Design (TIDD), PUC-SP.

Structure: follows the program's project outline (SANTAELLA, [n.d.]).

Area of concentration: Cognitive Processes and Digital Environments.

Research line: 4. Artificial Intelligence and Management, with secondary adherence to line 3, Human-Computer Interaction and Social Computing.

> **Responsible-use notice.** Automated image-based visual detection, subject to errors and not equivalent to operational, legal, safety or aeronautical validation. This research evaluates detectors, not sites: it does not produce, nor does it intend to produce, any validation of helipads for use, landing, delivery, or regulatory compliance.

Abstract

Object detectors trained on few examples are usually evaluated on a small validation set annotated by the team itself, and these metrics, on their own, do not characterize behavior in a real, unannotated region. In the case studied, the Helipad Detector, four experiments with mAP@50 from 0.993 to 0.994 produced, on the same 7,767 field region × tile pairs, detection rates from 9.6% to 21.0%. The dissertation asks how much of this difference is explained by data lineage, validation, and artifacts, and what the precision of the detections is according to sampled human review. It proposes a reproducible protocol with lineage audit, controlled comparison with spatial partitioning, and precision estimation by human review, treated as an act of image interpretation and not as ground truth. A study on reviewers' trust remains as an extension conditional on time and ethics approval.

Keywords: object detection; model evaluation; human visual reference; image interpretation; trust in automation; satellite imagery.

1. Research question

1.1 Context and problem

The Helipad Detector is a team-developed project, with a public repository ↗ and a record of each author's contributions (MINDFUL AI RESEARCH, 2026a). It uses YOLO networks trained for a single class (“helipad”), applied to satellite image crops (tiles of 256 × 256 pixels, zoom 19), with field evaluation in ten regions of Greater São Paulo, a scope that the research maintains.

The project originated as coursework for the Neural Networks course, in the 5th semester of the undergraduate program in Data Science and Artificial Intelligence at PUC-SP, taught by Prof. Rooney Ribeiro Albuquerque Coelho. The applicant has been part of the project since June 2026. The contribution that underpins this research is the automation of the collection and geocoding of tiles, the execution of experiments exp2, exp3, and exp4, and the field evaluation in the ten regions, expanded from the region explored by Pedro Vyctor Almeida, author of the original annotated dataset and of exp1. The project's other deliverables (interactive dashboard ↗, presentation ↗, and reports in Portuguese ↗ and English ↗) are in the Helipad Detector ↗ repository and are not part of this research (MINDFUL AI RESEARCH, 2026b, 2026c, 2026d, 2026e).

Ground truth is the independent confirmation, on site or from an official source, of what exists at the detected point. **Human visual reference** is the label that people assign from the same images: an act

of interpretation, subject to ambiguity, and not equivalent to ground truth. The project does not have the former at scale.

The experiments with YOLOv8n and YOLO11n are practically indistinguishable on validation (mAP@50 from 0.993 to 0.994; Table 1). On the same 7,767 region × *tile* pairs (5,540 distinct tiles) from ten regions, with a confidence threshold of 0.25, they produce detection rates from 9.64% to 21.00% (Table 2 and Figure 1): the highest is more than double the lowest.

Table 1. Performance on the validation set (best epoch by mAP@50–95)

Exp.	Model (epochs)	Data (train / val.)	Best epoch	Prec.	Rec.	mAP@50	mAP@50–95 best (last)
exp1	YOLOv8n (60)	Project-2 (178 / 34)†	54	1.000	0.963	0.994	0.881 (0.841)
exp2	YOLOv8n (100)	Project-2 (178 / 34)	81	0.982	0.971	0.993	0.888 (0.841)
exp3	YOLOv8n (100)	Project-1, with augmentation§ (356 / 34)	78	0.943	1.000	0.993	0.885 (0.841)
exp4	YOLO11n (100)	Project-2 (178 / 34)‡	54	0.970	0.948	0.993	0.877 (0.854)

Source: results.csv and args.yaml of the experiments, in the repository; seed 42 and 640 px images in all. †Attribution by documentary evidence, not by count record: the args.yaml and the preserved weights of exp1 (run "exp1-2", by Pedro Vyctor Almeida) point to version 2 of the dataset and are byte-for-byte identical to those of the applicant's copy; the training log saved in the notebook (116 training images and 33 validation images) is compatible with the original exp1 run, on version 1, of which only the args.yaml remains. Strong indication, to be confirmed in Stage 1. ‡Same data as exp2, but Ultralytics 8.4.163 (exp2: 8.4.98), which flagged as corrupted three training label files that mix polygons and boxes. §Fork of the dataset with data augmentation (rotation of ±15°, brightness, and flips).

For this reason, at present, no comparison among the repository's experiments constitutes a controlled experiment; the exp2 × exp4 pair is the closest to one, with the caveat of the Ultralytics version difference and the three flagged label files.

Table 2. Field detection (confidence threshold 0.25; 7,767 region × *tile* pairs, ten regions)

Experiment	Tiles with detection	Rate	Wilson 95% CI (naive)
exp1	749	9.64%	9.01–10.32%
exp3	1,389	17.88%	17.05–18.75%
exp4	1,423	18.32%	17.48–19.20%
exp2 (rerun)	1,631	21.00%	20.11–21.92%

Source: detection_summary_by_region*.json reports in the repository. The production run of exp2 (7,943 tiles) detected 1,677 (21.1%). The naive interval treats tiles as independent and therefore underestimates uncertainty, since neighboring tiles are spatially dependent.

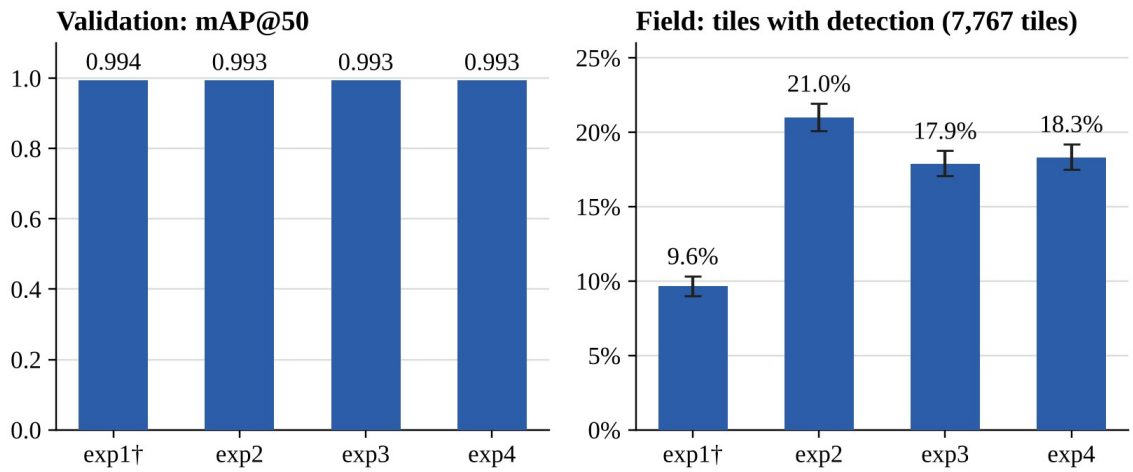


Figure 1. Same family of models, nearly identical validation metrics, and very different field detection rates. Error bars: 95% Wilson interval (naive). †Data version of exp1 attributed by documentary evidence (Table 1). Source: the author's own elaboration with repository data.

The detection rate is the fraction of tiles with at least one detection; it is neither precision nor recall, and without ground truth it is not known whether a higher rate means more helipads found or more false positives. The project has only case-by-case reviews, selected without random sampling (at Faria Lima, for example, three hits, two false positives, and one empty tile, due to a download failure, marked as a detection). Therefore, field precision and the share of the rate caused by defective tiles are currently unknown and will be estimated in Stages 1 and 3. An mAP close to 1 does not reveal this.

There is also a leakage risk, already partly measured. Comparing the annotated images, pixel by pixel, with the field *tiles* reconstructed from the repository (5,540 distinct *tiles*, in 7,767 region × *tile* pairs, since neighboring regions overlap), 125 of the 225 images with coordinates (87 of 177 in training, 25 of 34 in validation, and 13 of 14 in test) are copies of a field *tile*, which corresponds to 2.3% of the distinct *tiles*. The remaining 100 do not match any field *tile* and fall outside the boxes of the ten regions. Training has therefore already seen 87 *tiles* from the field, and the detection rate on those *tiles* tends to be inflated; the overall effect cannot be quantified without the detections per *tile*. There is no direct copy between validation or test and training, but 15 of the 34 validation images and 3 of the 14 test images are within one *tile* (about 70 m) of a training image, a neighborhood that will be treated as a risk, and not as confirmed leakage.

1.2 Central question and derived questions

Central question. How much of the field detection difference among YOLO detectors trained on few examples is explained by data lineage, validation choices, and data artifacts, and what is the precision of those detections according to sampled human review?

Derived questions.

- **SQ1 (lineage audit).** How much of the field differences stems from data lineage (version, partition, overlap with the field, defective *tiles*) and how much from architecture?
- **SQ2 (controlled comparison and spatial partition).** How much do best-epoch selection on the validation set itself and random partitioning overestimate performance in new regions?
- **SQ3 (sampled human review).** What is the field precision, by confidence band and by region, according to the human visual reference, and how reliable is that reference (agreement and proportion of “uncertain”)?

The influence of how confidence is presented on reviewers' decisions (SQ4, exploratory) is an extension conditional on time and ethics approval (section 7).

2. State of the art

Detection in aerial images. Object detection in remote sensing is reviewed by Cheng and Han (2016) and relies on large datasets, such as DOTA (XIA et al., 2018; DING et al., 2022). The YOLO family (REDMON et al., 2016; ULTRALYTICS, 2026), with COCO pretraining (LIN et al., 2014), is the basis of the models used, and Padilla et al. (2021) show that mAP depends on implementation conventions that must be declared. These works address large annotated datasets, not trusting a detector trained on few examples in an unannotated region.

Helipad detection. Helipad detection in aerial and satellite images has already been studied. Bitoun and Winkler (2020) present HelipadCat, a dataset of about 6,000 images in 12 categories, built from the FAA database, and a detection method with bounding-box annotations, Mask R-CNN, mAP evaluation, area and shadow filters, and map marker generation. Specht et al. (2021) train convolutional networks (ResNet101, Inception-V3, Xception, and EfficientNet-b0) to classify satellite images as helipad or not, on a dataset of 9,324 images, and report about 95% validation accuracy with EfficientNet-b0; the authors also record inaccuracies in the FAA 5010 database. Suprapto et al. (2021) use Tiny YOLO to detect helipads with an onboard camera, aimed at drone landing, a task distinct from this research. These works show that detecting helipads is not, by itself, the contribution here. The classification accuracy of Specht et al. (2021) is not comparable to the mAP of a detector, since tasks and metrics differ. Bitoun and Winkler (2020), in turn, use bounding boxes and mAP evaluation and are the closest antecedent; a numerical comparison with them would require verifying the compatibility of datasets, train and test splits, and evaluation criteria and, without that, remains methodological and contextual, without claiming performance superiority. The research is delimited by the evaluation of the detector in an unannotated region, with spatial partitioning and sampled human review.

Validity of evaluation. Kaufman et al. (2012) formalize data leakage. Roberts et al. (2017) and Ploton et al. (2020) show, for spatial data, that validation that respects spatial structure reveals worse performance than the apparent one; these results come from ecology, and their transfer to detectors on satellite images is a hypothesis to be tested (H2). Dietterich (1998) addresses tests for comparing models, and Brown, Cai, and DasGupta (2001), intervals for proportions, the basis of the Wilson intervals used in Table 2.

Calibration. Guo et al. (2017) show that modern networks tend to be poorly calibrated, in image classification; the calibration of detectors with little data will be investigated from precision by confidence band (Stage 3), since confidence is the main signal that reviewers receive.

Image interpretation and cognition. In the semiotics of the image, seeing is not recording a fact but interpreting a sign, and interpretation depends on conventions, repertoire, and context (SANTAELLA; NÖTH, 1998). A satellite image is a representation made by a sensor, at a given date and resolution, and using it to assert what exists on the ground requires inference. Distributed cognition (HUTCHINS, 1995) describes reasoning as a process that extends across people, instruments, and procedures. Both support treating the review of detections as a cognitive activity situated in a digital environment, and not as a neutral reading of an answer key.

Trust in automation and transparency. Parasuraman and Riley (1997) distinguish use, misuse, disuse, and abuse of automation; Lee and See (2004) address appropriate trust; Amershi et al. (2019) propose human-AI interaction guidelines. *Model Cards* (MITCHELL et al., 2019) and *Datasheets for Datasets* (GEBRU et al., 2021) propose reporting intended uses and limits, a practice that the project adopts.

Gap. In the preliminary review, restricted to the sources already consulted, no work was identified that combines, for a rare urban object in satellite images, controlled comparisons, spatial validation, and precision estimation by sampled human review that treats the human reference as interpretation, with agreement and uncertainty declared. The search was exploratory on the web, with combinations of the terms “helipad detection”, “satellite imagery”, “aerial images”, “deep learning”, and “YOLO”, and was not systematic or exhaustive. The systematic review (Stage 1) may change this conclusion.

3. Justifications

Theoretical and practical. The works reviewed address, separately, metrics, evaluation validity, calibration, and trust in automation. This case brings them together in a concrete problem and shows a gap between validation metric and field behavior that is not solved with more training. The resulting protocol (lineage audit, spatial partitioning, sampled human review, and audit log) can be reapplied to other detectors and regions and turns an uninterpreted detection rate into estimates with explicit uncertainty.

Fit with TIDD. The object of this research, in the area Cognitive Processes and Digital Environments, is not the detector itself but the production of evidence about it in a digital environment: people look at satellite images in an interface, assign labels (helipad, not helipad, uncertain), and, with them, decide whether to trust the system. Two cognitive phenomena are at play. First, the human visual reference is an act of interpretation: it depends on the reviewer's repertoire and on the image's resolution and date, and generates disagreement and hesitation that need to be measured, not eliminated. Second, the reviewer's trust in the detector is a human-automation interaction phenomenon (PARASURAMAN; RILEY, 1997; LEE; SEE, 2004), influenced by what the interface shows (AMERSHI et al., 2019). The fit is also in the syllabi of the program ↗: the area of concentration deals with "modeling of cognitive performances: perception, attention, inference", and line 4 describes the passage to "a world of probabilistic machines, implying distinct logics and risks", with attention to the ethical and social impacts of AI. A detector whose confidence is read by people is a concrete case of that world. For this reason, the research is situated in line 4, through the evidence produced about a probabilistic machine and the limits of automation, and, secondarily, in line 3, through people's interaction with AI systems and governance.

Social and ethical. A helipad detector may be misinterpreted as evidence that a location is safe, operational, or authorized. None of this is true. All outputs will be treated as analytical and non-authoritative, with the notice at the beginning of this document.

4. Objectives

General objective. To propose and test a reproducible protocol for evaluating visual helipad detectors on satellite images without ground truth at scale, treating human review as an interpretive reference, with uncertainty and agreement declared.

Specific objectives.

1. Reconstruct the lineage of the data and configurations of the existing experiments, verifying the overlap between training and field and the defective *tiles* (SQ1).
2. Carry out controlled comparisons between YOLOv8n and YOLO11n and quantify the optimistic bias of best-epoch selection and of random partitioning, relative to a spatial partition by region (SQ1 and SQ2).
3. Estimate field precision, by confidence band and by region, through blind sampled human review, performed by two reviewers (the applicant and Pedro Vyctor Almeida), reporting inter-reviewer agreement and the handling of "uncertain" (SQ3).
4. Document the system and data with *Model Card*, *Data Card*, and an auditable review protocol.

Expected results. Lineage report; performance estimates with uncertainty intervals, by partition and by region; precision estimate by sampled human review; *Model Card*, *Data Card*, and review protocol; at least one paper. Operational, legal, or aeronautical validation of any detected location is not expected, nor will it be claimed.

5. Hypotheses

Hypothesis	Test	Refutation
H1. With practically equal mAP on validation, the field detection rate varies more across data versions and artifacts (such as empty *tiles*) than across architectures trained on the same data.	Stages 1 and 2	The difference between architectures, with identical data and several seeds, is larger than the difference between data versions.
H2. Best-epoch selection on the same validation set and random partitioning overestimate performance; under spatial partitioning, metrics drop measurably.	Stage 2	The difference between random and spatial partitioning falls within the uncertainty interval (*bootstrap*).
H3. Field precision, estimated by human review, is lower than validation precision; variation across regions will only be described.	Stage 3	The 95% interval of field precision is compatible with that of validation. With about 30 items per region, differences between regions will not be tested, only described.

Extension (H4, exploratory). Showing calibrated confidence and examples of typical errors reduces the acceptance of false positives by reviewers, compared with showing only the box and raw confidence. Its expected size will not allow strong inference.

6. Theoretical framework

The research rests on four axes, chosen because they stem from the gap between validation metric and field behavior. The first is **detection as probabilistic prediction**, summarized by precision, recall, and mAP (LIN et al., 2014; PADILLA et al., 2021). The second is **evaluation validity**: leakage (KAUFMAN et al., 2012), spatial dependence (ROBERTS et al., 2017; PLOTON et al., 2020), and model selection on the reporting set itself are sources of optimism. The third is **image interpretation and the notion of evidence** (6.1). The fourth is **trust, uncertainty, and responsible AI**: calibration (GUO et al., 2017), appropriate trust (PARASURAMAN; RILEY, 1997; LEE; SEE, 2004; AMERSHI et al., 2019), transparency (MITCHELL et al., 2019; GEBRU et al., 2021), and protection of personal data (BRASIL, 2018), since images of rooftops may allow inferences about people and buildings.

6.1 Evidence when there is no ground truth

Interpreting an image is a situated act: the same crop can be read in different ways depending on the viewer's repertoire and on the image's resolution and date (SANTAELLA; NÖTH, 1998), and the reviewer's judgment is formed in a system that includes interface, criteria guide, and model output (HUTCHINS, 1995). For this reason, the human label is not treated as an answer key. **Evidence**, here, is any statement about the detector's field performance that has traceable provenance (lineage), a reproducible procedure, declared uncertainty, and an indication of the level at which it sits (Table 3). A statement without a declared level is not evidence. Disagreement between reviewers and the “uncertain” category are study data, not noise to be discarded.

Table 3. Evidence levels adopted in the research

Level	What it is	What it allows one to claim	What it does not allow one to claim
1. Validation metric	Precision, recall, and mAP on the team-annotated set.	Performance on the small validation set.	Performance in unannotated regions.
2. Field detection rate	Fraction of *tiles* with at least one detection.	Compare detectors on the same *tiles*.	Precision, recall, or number of helipads.
3. Sampled human visual reference	Labels from two reviewers on a stratified sample, with agreement and "uncertain".	Estimate precision by stratum, with explicit uncertainty.	Ground truth; large-scale recall.
4. Ground truth	Independent confirmation, on site or from an official source.	Out of scope.	Will not be produced or claimed.

7. Methodology

Type of research. Applied and empirical, predominantly quantitative (computational experiments and statistical estimation), with an interpretive component: human review as the production of a visual reference.

Starting material and authorization. The repository 3-project-ai-ml-yolo-helipad_detector ↗, from the organization Mindful-AI-Research ↗, created by the applicant, is public and gathers experiments exp1 to exp4, the field reports, and the documentation. The use of the repository and data in this research has the authorization of Pedro Vyctor Almeida, co-author of the original annotated dataset, recorded in an authorization form ↗ signed by Pedro Vyctor Almeida and the applicant. The data are small (226 images in version 2: 178 training, 34 validation, and 14 test), with images from a single provider and date. The provider's terms of use for derived datasets and redistribution will be verified in Stage 1, before any data publication. The project presentation ↗ summarizes the work and the method.

Table 4. Dependencies and fallback plans

Dependency	Risk	Plan B
exp1 data version	No count record matching the preserved metrics; attribution to version 2 by documentary evidence.	If Stage 1 does not confirm it, exp1 leaves the comparisons and remains only as a descriptive observation; SQ1 and SQ2 do not depend on it (Stage 2 retrains).
Flagged labels in exp4	Three training label files mix polygons and boxes; Ultralytics 8.4.163 flags them as corrupted.	Stage 1 standardizes the labels; Stage 2 retrains both models on the same version, with verified data; if unfeasible, exp2 × exp4 stands with a caveat.
Authorization of use	Pedro Vyctor Almeida withdraws the signed authorization, or the provider's terms (Esri World Imagery, third party) restrict the use or redistribution of the images and of the derived dataset.	Restrict to the material authored by the applicant (collection, geocoding, exp2 to exp4, field reports), re-annotating as needed.
Second reviewer	Pedro Vyctor Almeida accepted (signed form); the risk is unavailability throughout the project.	Invite a substitute with the same profile; without one, report with a single reviewer and declare the limitation.

Stage 1. Lineage audit and reproducibility. Reconstruct, for each experiment, the data version, the partition, the configuration, and the seed, confirming the exp1 data version, standardizing the labels (most files contain polygons, which detection training converts into boxes; three mix polygons and boxes), and verifying the effect of the Ultralytics version. Extend the training × field cross-check already done (*tile* copies and neighborhood) to the *tiles* missing from Faria Lima and to the detections per *tile* of each experiment to measure the overlap, measure the share of empty or corrupted *tiles* sent to inference, and pin the versions of data, code, and dependencies. In parallel, carry out the systematic literature review, including image semiotics and cognition.

Stage 2. Controlled comparisons and spatial partitioning. Train YOLOv8n and YOLO11n with the same data, the same partition, and the same epochs, with at least three seeds, and compare with tests for paired models, such as McNemar and 5×2 cross-validation (DIETTERICH, 1998), and Wilson intervals (BROWN; CAI; DASGUPTA, 2001). Then, train and evaluate with partitioning by region (leaving one out in each round, when feasible), compare with random partitioning, and quantify the difference. Since neighboring regions share tiles (5,540 distinct in 7,767 pairs), the partition will be made over distinct tiles, removing from training the tiles shared with the held-out region. Baixada Santista is a candidate for the held-out region, provided its absence from training is confirmed by coordinates; Faria Lima only qualifies if the two training images that already coincide with its tiles are excluded. The epoch will be chosen on a set separate from the one used for reporting.

Stage 3. Sampled human review in the field. Draw, stratified by confidence band and by region, detections from the Stage 2 model(s) over the 7,767 common region × *tile* pairs, plus a sample of *tiles* without detection, composed mainly of tiles on which the experiments disagree, plus a small random baseline sample, to estimate omissions, and produce the human visual reference according to Table 5, with two reviewers: the applicant, Fabiana Rotella Campaner, and Pedro Vyctor Almeida. Precision by stratum is estimated with Wilson intervals and spatial block *bootstrap*; the confidence threshold is varied to show how precision and coverage change, and precision by band feeds the reliability diagram (GUO et al., 2017). The result is human visual reference, and not ground truth.

Table 5. Human review protocol (Stage 3)

Item	Definition
Reviewers	Two: the applicant and Pedro Vyctor Almeida, who agreed to participate. He annotated the original dataset and trained exp1, and is therefore not independent of the project; the limitation is declared and a subsample of about 100 items will also be reviewed by a person external to the team, with kappa computed against both reviewers.
Independence and blinding	Each reviewer works alone. The queue is generated by a script with a fixed seed, in random order, mixing detections and *tiles* without detection; reviewers do not see the confidence, the source experiment, or the stratum. Since both know the detector, and the second reviewer annotated the training data (his criteria may coincide with those of the training label), the main analysis is repeated with each reviewer separately (sensitivity).
Categories and pilot	Helipad, not helipad, and uncertain, with visual criteria in a written guide. Pilot of about 50 items to adjust the guide and sample and compute agreement (Cohen's kappa).
Sample (order of magnitude)	Three confidence bands (0.25–0.50; 0.50–0.75; ≥ 0.75), with about 100 detections per band, drawn with a minimum quota per region; strata with fewer than about 100 are reviewed in full. Plus about 100 *tiles* without detection, composed mainly of tiles on which the experiments disagree (detected by one model and not by another, or with a detection just below the 0.25 threshold), plus a small random baseline sample. Total on the order of 400 items per reviewer, to be adjusted by the pilot.
Power for rare strata	With n ≈ 100 per band, the half-width of the Wilson interval is about ±10 percentage points (p = 0.5), and comparing two bands has about 80% power for differences of 20 points (50% versus 70%). By region (n ≈ 30), the half-width reaches ±17 points, and results are descriptive. With few expected positives, only the upper bound is reported (rule of three: zero in 100 implies about 3%). Since spatial dependence widens the intervals, block *bootstrap* is used. The sample of disagreement tiles is not representative: it shows types of omission, and only the random baseline sample estimates the omission rate.
"Uncertain" and disagreements	"Uncertain" is its own category, never silently merged: precision is reported with "uncertain" as not helipad, as helipad, and excluded, together with its proportion per stratum. Kappa is computed before any resolution; disagreements are discussed in a recorded session and, if they persist, the item stays "uncertain".

Stage 4. Documentation and audit. Produce *Model Card*, *Data Card*, review protocol, and audit log schema, with versioned artifacts.

Conditional extension (H4, SQ4). Compare two interfaces (box and raw confidence *versus* calibrated confidence with examples of typical errors), measuring acceptance of false positives and decision time. It requires ethics approval: the project will be submitted in semester 2 and the

extension will only be carried out if approval arrives before semester 4 and time allows; otherwise, it remains as future work, without prejudice to the central scope. The design and analysis methods of this study will be deepened in a human-computer interaction course (section 8) and in the bibliography to be consulted.

Ethics, privacy, and limitations. No publication will identify specific people or buildings, and results will be aggregated by region; the need to process personal data will be assessed in light of the LGPD (BRASIL, 2018). Anticipated limitations: small and geographically heterogeneous dataset; a single provider (Esri World Imagery) and image date; absence of ground truth at scale; cost of human review; ambiguity of visual interpretation; and limited power in rare strata.

8. Timeline (24 months, within the program's 18 to 30)

Semester	Activities	Deliverables
1 (months 1–6)	Courses: Epistemology and Research Methodology (EMP, mandatory) and Fundamentals of Artificial Intelligence (FIA, line 4, module 1); foreign-language proficiency, within 6 months of enrollment (call notice, item 6.4); systematic review; Stage 1; design of the review protocol.	Lineage report; review protocol (v1)
2 (months 7–12)	Courses: two electives, candidates Human-Computer Interaction and User Experience (line 3) and Theory of Signs, Symbols, and Codes (line 4); Stage 2; review pilot; submission to the ethics committee (extension).	Results of SQ1 and SQ2; pilot report
3 (months 13–18)	Qualifying exam (semester 3); Stage 3; decision on the extension, with ethics approval in hand.	Qualifying text; results of SQ3
4 (months 19–24)	Stage 4; extension with reviewers, if approved and time allows; writing; paper; defense.	Dissertation; at least one paper submitted

The TIDD master's program lasts 18 to 30 months (program page ↗); 24 are proposed. The candidate courses depend on what is offered, since the program states that the syllabus of each receives, every semester, a content specification; the qualifying exam follows the program's regulations.

Preliminary bibliography

Preliminary list, according to ABNT standards, of the works consulted in preparing this project, followed by the works to be consulted. It will be supplemented during the systematic literature review.

AMERSHI, S. et al. Guidelines for human-AI interaction. *In*: PROCEEDINGS OF THE 2019 CHI CONFERENCE ON HUMAN FACTORS IN COMPUTING SYSTEMS, 2019, Glasgow. New York: ACM, 2019. p. 1–13.

BITOUN, J.; WINKLER, S. HelipadCat: categorised helipad image dataset and detection method. In: IEEE REGION 10 CONFERENCE (TENCON), 2020, Osaka, Japan. *Proceedings* [...]. [S. l.]: IEEE, 2020. p. 685–689.

BRASIL. *Law No. 13,709, of 14 August 2018*. General Personal Data Protection Law (LGPD). Brasília, DF: Presidency of the Republic, 2018. Available at: https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/lei/l13709.htm ↗. Accessed: 13 Oct. 2026.

BROWN, L. D.; CAI, T. T.; DASGUPTA, A. Interval estimation for a binomial proportion. *Statistical Science*, v. 16, n. 2, p. 101–133, 2001. DOI 10.1214/ss/1009213286 ↗.

CHENG, G.; HAN, J. A survey on object detection in optical remote sensing images. *ISPRS Journal of Photogrammetry and Remote Sensing*, v. 117, p. 11–28, 2016.

DIETTERICH, T. G. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, v. 10, n. 7, p. 1895–1923, 1998.

DING, J. et al. Object detection in aerial images: a large-scale benchmark and challenges. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, v. 44, n. 11, p. 7778–7796, 2022.

GEBRU, T. et al. Datasheets for datasets. *Communications of the ACM*, v. 64, n. 12, p. 86–92, 2021.

GUO, C. et al. On calibration of modern neural networks. *In*: INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML), 34., 2017, Sydney. [S. l.]: PMLR, 2017. *Proceedings of Machine Learning Research*, v. 70, p. 1321–1330, 2017.

HUTCHINS, E. *Cognition in the wild*. Cambridge, MA: MIT Press, 1995.

KAUFMAN, S. et al. Leakage in data mining: formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, v. 6, n. 4, art. 15, 2012.

LEE, J. D.; SEE, K. A. Trust in automation: designing for appropriate reliance. *Human Factors*, v. 46, n. 1, p. 50–80, 2004.

LIN, T.-Y. et al. *Microsoft COCO: common objects in context*. arXiv:1405.0312 ↗, 2014. Available at: https://arxiv.org/abs/1405.0312 ↗. Accessed: 13 Oct. 2026.

MINDFUL AI RESEARCH. Helipad Detector: 3-project-ai-ml-yolo-helipad_detector. Repository. GitHub, 2026a. Available at: https://github.com/Mindful-AI-Research/3-project-ai-ml-yolo-helipad_detector ↗. Accessed: 13 Oct. 2026.

MINDFUL AI RESEARCH. Bilingual dashboard: Streamlit. Helipad Detector project material. 2026b. Available at: https://helipad-detector-sp.streamlit.app/ ↗. Accessed: 13 Oct. 2026.

MINDFUL AI RESEARCH. Full Data Analysis Report: English. Helipad Detector project material. 2026c. Available at: https://github.com/Mindful-AI-Research/3-project-ai-ml-yolo-helipad_detector/blob/db052a43d2b02254237735716dc841a3611d9e98/reports/helipad_detector_full_report/%F0%9F%87%AC%F0%9F%87%A7Helipad_Detector_Full_Report.pdf ↗. Accessed: 13 Oct. 2026.

MINDFUL AI RESEARCH. Full data analysis report: Portuguese. Helipad Detector project material. 2026d. Available at: https://github.com/Mindful-AI-Research/3-project-ai-ml-yolo-helipad_detector/blob/3849b944fbfbf2df224fbd7e636751872c35258a/reports/helipad_detector_full_report/%F0%9F%87%A7%F0%9F%87%B7Helipad_Detector_Relatorio_Completo.pdf ↗. Accessed: 13 Oct. 2026.

MINDFUL AI RESEARCH. Helipad Detector project presentation. Helipad Detector project material. 2026e. Available at: https://timely-brioche-b6c92f.netlify.app/ ↗. Accessed: 13 Oct. 2026.

MITCHELL, M. et al. Model cards for model reporting. *In*: PROCEEDINGS OF THE CONFERENCE ON FAIRNESS, ACCOUNTABILITY, AND TRANSPARENCY, 2019, Atlanta. New York: ACM, 2019. p. 220–229.

PADILLA, R. et al. A comparative analysis of object detection metrics with a companion open-source toolkit. *Electronics*, v. 10, n. 3, art. 279, 2021.

PARASURAMAN, R.; RILEY, V. Humans and automation: use, misuse, disuse, abuse. *Human Factors*, v. 39, n. 2, p. 230–253, 1997.

PLOTON, P. et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, v. 11, art. 4540, 2020. DOI 10.1038/s41467-020-18321-y ↗.

REDMON, J. et al. You only look once: unified, real-time object detection. *In*: IEEE CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR), 2016, Las Vegas. [S. l.]: IEEE, 2016. p. 779–788.

ROBERTS, D. R. et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, v. 40, n. 8, p. 913–929, 2017.

SANTAELLA, L. *Roteiro de projeto*. São Paulo: PEPG in Intelligence Technologies and Digital Design, PUC-SP, [n.d.]. Available at: https://www.pucsp.br/pos-graduacao/mestrado-e-doutorado/tecnologias-da-inteligencia-e-design-digital ↗. Accessed: 13 Oct. 2026.

SANTAELLA, L.; NÖTH, W. *Imagem*: cognição, semiótica, mídia. São Paulo: Iluminuras, 1998.

SPECHT, D.; RASOOL, G.; BOUAYNAYA, N.; JOHNSON, C. Intelligent helipad detection from satellite imagery. In: VERTICAL FLIGHT SOCIETY ANNUAL FORUM AND TECHNOLOGY DISPLAY, 77., 2021, online. *Proceedings* [...]. [S. l.]: Vertical Flight Society, 2021. DOI 10.4050/F-0077-2021-16856.

SUPRAPTO, B. Y. et al. The detection system of helipad for unmanned aerial vehicle landing using YOLO algorithm. *Jurnal Ilmiah Teknik Elektro Komputer dan Informatika* (JITEKI), v. 7, n. 2, p. 193–206, 2021. DOI 10.26555/jiteki.v7i2.20684.

ULTRALYTICS. *Ultralytics YOLO*. Software. [S. l.]: Ultralytics, 2026. Available at: https://github.com/ultralytics/ultralytics ↗. Accessed: 13 Oct. 2026.

XIA, G.-S. et al. DOTA: a large-scale dataset for object detection in aerial images. *In*: IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR), 2018, Salt Lake City. [S. l.]: IEEE, 2018. p. 3974–3983.

Works to consult (bibliography of the program's courses, TIDD page ↗; not yet consulted for this project):

CHALMERS, A. F. *O que é ciência, afinal?* São Paulo: Brasiliense, 1993.

LAZAR, J.; FENG, J.; HOCHHEISER, H. *Research methods in human-computer interaction*. 2nd ed. Cambridge, MA: Morgan Kaufmann, 2017.

RUSSELL, S.; NORVIG, P. *Inteligência artificial*. 3rd ed. Rio de Janeiro: Elsevier, 2013.

SANTAELLA, L. *Comunicação & pesquisa*. 2nd ed. São Paulo: Hacker; Bluecom, 2010.

SANTAELLA, L.; VIEIRA, J. A. *Metaciência*. São Paulo: Mérito, 2008.